

Civil Engineering

Vectors, Matrices and Data Analysis

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\vec{R} = \vec{A} + \vec{B}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$AX = B$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\det(A) = ad - bc$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Section B

5. Resolve forces into components. **(10 marks)**
6. Solve simultaneous equations using matrices. **(10 marks)**
7. Calculate standard deviation of concrete test results. **(10 marks)**
8. Create one realistic civil engineering problem based on this topic and solve it fully. **(20 marks)**